NETFLIX RETENTION INTELLIGENCE REPORT

AI-DRIVEN CUSTOMER CHURN PREDICTION & RETENTION STRATEGY

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1. Executive Summary

This intelligence audit analyzes a portfolio of **800 active subscribers** representing **$12,000.00** in monthly recurring revenue (MRR). Our calibrated machine-learning pipeline identifies **120 subscribers (15.0%)** in the High-Risk category, placing **$1,800.00 MRR directly at risk** of cancellation. Deploying targeted, deterministic retention interventions across the portfolio is projected to generate **$2,400.00** in expected incremental net contribution benefit against a campaign investment of **$500.00** (Portfolio ROI: **480.0%**).

2. Business Problem & Objectives

Subscription video-on-demand businesses face subscriber churn despite continued customer acquisition spending. The objective of this AI system is to identify early behavioral indicators prior to cancellation, quantify financial exposure, distinguish prospective active customers from historical churners, and match each subscriber with targeted, advisory retention interventions.

3. Dataset Description & Data Quality Audit

Audit Dimension	Observed Metric	Quality Assessment
Total Analyzed Records	1,000 records	Complete synthetic cohort with verified schema
Active Prospective vs Churned	800 Active / 200 Historical	Strict isolation of active prospective subscribers
Observed Churn Rate	20.0%	Realistic SVOD annual/quarterly churn baseline
Data Hygiene & Missing Values	0 critical missing values / 0 duplicates	Robust imputation & scaling pipeline active

4. Machine Learning Methodology & Performance

Models were evaluated using a stratified 80/20 train/test split. Candidate models: Logistic Regression (balanced) and Random Forest (balanced). The selected model for inference is **Logistic Regression**, chosen to optimize sensitivity towards identifying at-risk accounts while maintaining well-calibrated probabilities.

Candidate Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.780	0.650	0.720	0.680	0.840

5. Predicted Risk Segment Distribution

Risk Tier	Threshold Criteria	Subscribers	% of Base	Monthly Revenue	MRR at Risk
High Risk	P(Churn) ≥ 70%	120	15.0%	$2,400.00	**$1,800.00**
Medium Risk	40% ≤ P(Churn) < 70%	280	35.0%	-	$1,400.00
Low Risk	P(Churn) < 40%	400	50.0%	-	Nominal

Total Active Portfolio	All Active Accounts	**800**	100.0%	**$12,000.00**	**$3,200.00**

6. Important Churn-Associated Indicators

The following table ranks domain behavioral features by statistical association with cancellation. *Note: Predictive importance denotes statistical correlation within historical observations, not proven causation.*

Behavioral Domain Driver	Importance Share (%)	Observed Business Implication
Watch Hours Drop	35.0%	Primary early warning indicator of subscriber disengagement
Inactivity	25.0%	Primary early warning indicator of subscriber disengagement

7. Sample High-Risk Subscribers Requiring Immediate Review

Customer ID	Risk Prob	Plan	Watch Drop	Inactivity	Recommended Advisory Action
NFLX-100	85.0%	Premium	50.0%	18 days	Targeted Win-Back

8. Retention Intervention Prioritization & Expected ROI

Calculations assume a 75% contribution margin over a 6-month evaluation horizon, and a 20% relative retention uplift. *Financial figures represent modeled business scenarios and must be validated through controlled holdouts.*

Rank	Recommended Campaign	Eligible Subs	Total Cost	Expected Net Benefit	ROI (%)
1	**Frictionless Payment Resolution**	45	$112.50	**$450.00**	400.0%

9. Recommended Next Steps for Management

1. **Deploy Payment Recovery Workflow:** Prioritize urgent automated billing retries and card-update prompts (Rank 1 ROI).
2. **Implement Randomized A/B Holdout (15%):** Never roll out blanket interventions without uncontacted holdout controls to measure real incremental uplift.
3. **Prevent Dilutive Discounting:** Avoid offering price reductions to low-risk customers whose organic retention is high.
4. **Human Review:** Ensure high-touch customer care outreach is confirmed by authorized team members prior to deployment.

10. Limitations & Disclaimers

Notice of Independence: *This is an independent educational prototype developed for simulation purposes. It is not affiliated with, endorsed by, or using internal systems/data of Netflix, Inc.*
Causal Limitation: *Predictive machine learning models identify statistical correlates. They do not demonstrate direct causation. Financial projections represent sensitivity scenarios rather than guaranteed revenue.*